Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Let d be the perpendicular distance from the centre of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ to the tangent drawn at a point P on the ellipse. If F_1 and F_2 be the foci of the ellipse, then $\left(\mathrm{PF}_1-\mathrm{PF}_2\right)^2=$

(A) $4a^2\left(1-\frac{b^2}{d^2}\right)$

(B) $\mathrm{a}^2\left(1-\frac{\mathrm{b}^2}{\mathrm{d}^2}\right)$

(C) $4b^2\left(1-\frac{a^2}{d^2}\right)$

(D) $b^2\left(1-\frac{a^2}{d^2}\right)$

Q.2

If a variable point P on an ellipse of eccentricity e is joined to the foci S_1 and S_2 then the incentre of the triangle $\mathrm{PS}_1\ \mathrm{S}_2$ lies on

(A) The major axis of the ellipse

(B) The circle with radius e

(C) Another ellipse of eccentricity $\sqrt{\frac{3+\mathrm{e}^2}{4}}$

(D) None of these

Q.3

The equation of the ellipse with its centre at $(1,2)$, focus at $(6,2)$ and passing through the point $(4,6)$ is $\frac{(x-1)^2}{a^2}+\frac{(y-2)^2}{b^2}=1$, then

(A) $a^2=1, b^2=25$

(B) $a^2=25, b^2=20$

(C) $a^2=20, b^2=25$

(D) None of these

Q.4

An ellipse has OB as semi minor axis, F and F its focii and the angle FBF' is a right angle. Then the eccentricity of the ellipse is

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(A) $\frac{1}{\sqrt{2}}$

(B) $\frac{1}{2}$

(C) $\frac{1}{4}$

(D) $\frac{1}{\sqrt{3}}$

Q.5

If $P \equiv (x, y), F_1 \equiv (3, 0), F_2 \equiv (-3, 0)$ and $16x^2 + 25y^2 = 400$, then $PF_1 + PF_2$ equals

(A) 8

(B) 6

(C) 10

(D) 12

Q.6

If a point $P(x, y)$ moves along the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ and if C is the centre of the ellipse, then, $4\max\{CP\} + 5\min\{CP\} =$

(A) 25

(B) 40

(C) 45

(D) 54

Q.7

If the angle between the straight lines joining the foci to an extremity of minor axis in an ellipse be 90°; then the eccentricity of the ellipse is

(A) $\frac{1}{2}$

(B) $\frac{1}{\sqrt{3}}$

(C) $\frac{1}{\sqrt{2}}$

(D) $\frac{1}{3}$

Q.8

The curve represented by $x = 2(\cos t + \sin t),, y = 5(\cos t - \sin t)$ is

(A) a circle

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(B) a parabola

(C) an ellipse

(D) a hyperbola

Q.9

If the focal distance of an end of the minor axis of any ellipse (referred to its axis as the axes of x and respectively) is k and the distance between the foci is $2\ \mathrm{h}$, then its equation is

(A) $\frac{x^2}{k^2}+\frac{y^2}{k^2+h^2}=1$

(B) $\frac{x^2}{k^2}+\frac{y^2}{h^2-k^2}=1$

(C) $\frac{x^2}{k^2}+\frac{y^2}{k^2-h^2}=1$

(D) $\frac{x^2}{k^2}+\frac{y^2}{h^2}=1$

Q.10

If the latus rectum of an ellipse is equal to one half its minor axis, what is the eccentricity of the ellipse?

(A) $\frac{1}{2}$

(B) $\frac{\sqrt{3}}{2}$

(C) $\frac{3}{4}$

(D) $\frac{\sqrt{15}}{4}$

Q.11

The sum of the focal distances of a point on the ellipse $\frac{x^2}{4}+\frac{y^2}{9}=1$ is:

(A) 4 units

(B) 6 units

(C) 8 units

(D) 10 units

Q.12

A point on the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ at a distance equal to the mean of the lengths of the semimajor axis and semi-minor axis from the centre is

(A) $\left(\frac{2\sqrt{91}}{7},\frac{3\sqrt{105}}{14}\right)$

(B) $\left(\frac{2\sqrt{91}}{7},-\frac{3\sqrt{105}}{14}\right)$

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(C) $\left(\frac{2\sqrt{105}}{7}, \frac{3\sqrt{91}}{14}\right)$

(D) $\left(-\frac{2\sqrt{105}}{7}, -\frac{3\sqrt{91}}{14}\right)$

Q.13

Which one of the following is correct? The eccentricity of the conic

$$\frac{x^2}{a^2+\lambda}+\frac{y^2}{b^2+\lambda}=1, (\lambda \geq 0)$$

(A) increases with increase in λ

(B) decreases with increase in λ

(C) does not change with λ

(D) None of the above

Q.14

The distance between the foci of an ellipse is 10 and its latus rectum is 15 . Its equation referred to its axes as axes of coordinates is

(A) $3x^2+4y^2=300$

(B) $2x^2+y^2=50$

(C) $10x^2+15y^2=300$

(D) None of these

Q.15

If the axes of an ellipse coincides with the co-ordiante axes and it passes through the point $(4,-1)$ and touches the line $x+4y-10=0$ then the eq. is

(A) $\frac{x^2}{80}+\frac{y^2}{5/4}=1$

(B) $\frac{x^2}{20}+\frac{y^2}{5}=1$

(C) $\frac{x^2}{100}+\frac{y^2}{5}=1$

(D) Both (a) and (b)

Q.16

An ellipse has OB as semi minor axis, F and F' its focii and the angle FBF' is a right angle. Then the eccentricity of the ellipse is

(A) $\frac{1}{4}$

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(B) $\frac{1}{\sqrt{3}}$

(C) $\frac{1}{\sqrt{2}}$

(D) $\frac{1}{2}$

Q.17

S and S' are the foci of an ellipse and B is an end of the minor axis. If $\mathrm{SS}'\mathrm{P}$ is an equilateral triangle, the eccentricity of the ellipse is

(A) $\frac{1}{4}$

(B) $\frac{1}{3}$

(C) $\frac{1}{2}$

(D) $\frac{2}{3}$

Q.18

A man running round a racecourse notes that the sum of the distances of two flag-posts from him is always 10 meters and the distance between the flag-posts is 8 meters. The area of the path enclosed in square meters is

(A) 15π

(B) 12π

(C) 18π

(D) 8π

Q.19

If $3x + 4y = 12\sqrt{2}$ is a tangent to the ellipse $x^2/a^2 + y^2/9 = 1$, for some a € R then the distance between the foci of the ellipse is:

(A) $2\sqrt{5}$

(B) $2\sqrt{7}$

(C) $2\sqrt{2}$

(D) 4

Q.20

Equation of the ellipse whose focus is $(6, 7)$, directrix is $x + y + 2 = 0$ and $e = \frac{1}{\sqrt{3}}$ is

(A) $5x^2 + 2xy + 5y^2 - 76x - 88y + 506 = 0$

(B) $5x^2 - 2xy + 5y^2 - 76x - 88y + 506 = 0$

(C) $5x^2 - 2xy + 5y^2 + 76x + 88y + 506 = 0$

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(D) None of these

Answers & Solutions

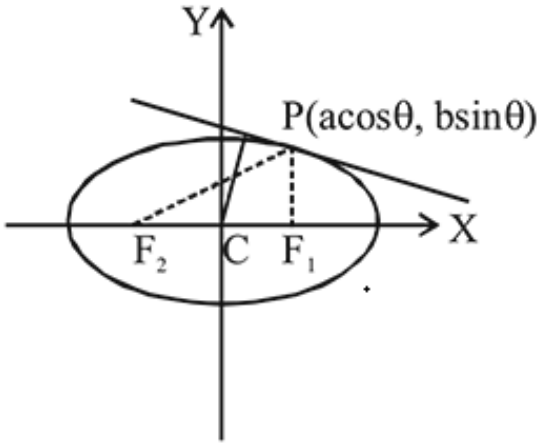
Q.1 Answer:

$4a^2\left(1-\frac{b^2}{d^2}\right)$

Solution:

Let the point P **be** $(\mathrm{a}\cos\theta, \mathrm{b}\sin\theta)$

The equation of tangent at P **is**



$\frac{\mathrm{x}\cos\theta}{\mathrm{a}}+\frac{\mathrm{y}\sin\theta}{\mathrm{b}}=1$

If d **be the length of perpendicular from the centre** $\mathrm{C}(0,0)$ **of the ellipse to the tangent given by** (1)

then $\mathrm{d}=\frac{1}{\sqrt{\frac{\cos^2\theta}{\mathrm{a}^2}+\frac{\sin^2\theta}{\mathrm{b}^2}}}$

$\Rightarrow \frac{1}{\mathrm{d}^2}=\frac{\cos^2\theta}{\mathrm{a}^2}+\frac{\sin^2\theta}{\mathrm{b}^2}$

$\Rightarrow \frac{\mathrm{b}^2}{\mathrm{d}^2}=\frac{\mathrm{b}^2}{\mathrm{a}^2}\cos^2\theta+1-\cos^2\theta$

$\Rightarrow 1-\frac{b^2}{d^2}=\left(1-\frac{b^2}{a^2}\right)\cos^2\theta=e^2\cos^2\theta \ldots(2)$

Now, $(\mathrm{PF}_1-\mathrm{PF}_2)^2=(2\mathrm{ae}\cos\theta)^2$

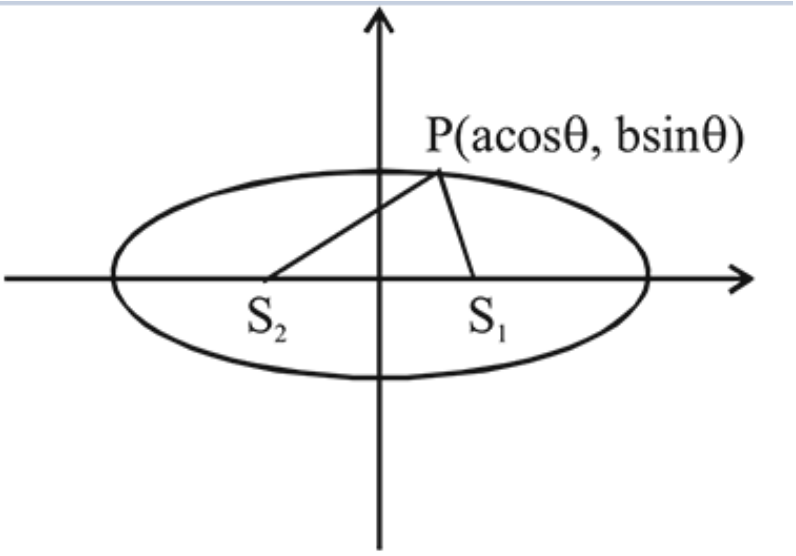
$=4a^2e^2\cos^2\theta=4a^2\left(1-\frac{b^2}{d^2}\right)$

Q.2 Answer:

Another ellipse of eccentricity $\sqrt{\frac{3+\mathrm{e}^2}{4}}$

Solution:

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Let the ellipse be $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1 \quad \ldots.$

Then $e^2=1-\frac{b^2}{a^2} \quad \ldots$

Let a point P **on** (1) **be** $(a\cos\theta, b\sin\theta)$.

The coordinates of foci are $S_1(ae,0)$ **and** $S_2(-ae,0)$**. Hence,** $S_1P=a(1-e\cos\theta)$

$S_2P=a(1+e\cos\theta)$ **and** $S_1\,S_2=2ae$

If (h,k) **be the coordinates of in centre then** $h=\frac{2ae\times a\cos\theta+a(1-e\cos\theta)\times -ae+a(1+e\cos\theta)\times ae}{2ae+a(1-e\cos\theta)+a(1+e\cos\theta)}$

$=\frac{2ae\cos\theta}{1+e}$

$k=\frac{be\sin\theta}{1+e}$

Squaring and adding $(3)\&(4)$ **we have,**

$\frac{h^2}{4a^2}+\frac{k^2}{b^2}=\left(\frac{e}{1+e}\right)^2$

$\therefore$ **The locus of the point** (h,k) **is** $\frac{x^2}{4a^2\lambda^2}+\frac{y^2}{b^2\lambda^2}=1$**, where** $\lambda=\frac{e}{1+e}$

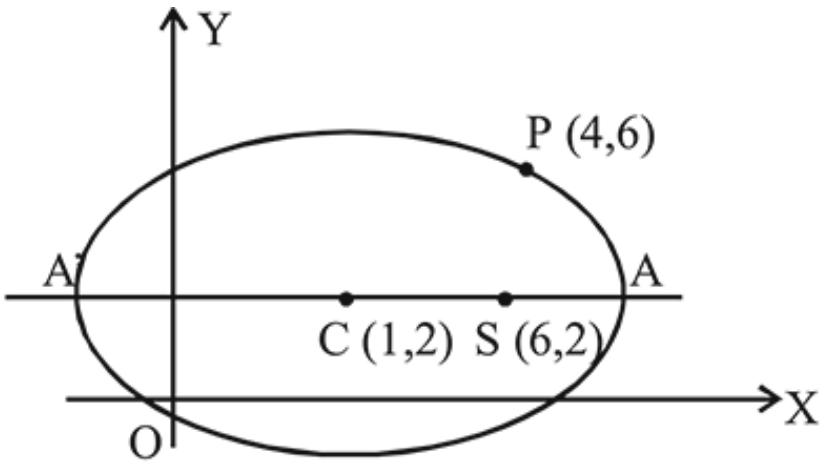
Which is another ellipse with eccentricity $=\sqrt{1-\frac{b^2}{4a^2}}=\sqrt{\frac{3+e^2}{4}}$

Q.3 Answer:

None of these

Solution:

Centre is $(1,2)$ **and focus is** $(6,2)$**, hence the line joining the centre** C **and the focus** S **(i..e., the axis) is parallel to x-axis. Therefore, the equation must be of the form**



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$\frac{(x-1)^2}{a^2}+\frac{(y-2)^2}{b^2}=1$

The distance between the centre and the focus $\mathrm{CS}=\mathrm{ae}=6-1=5$

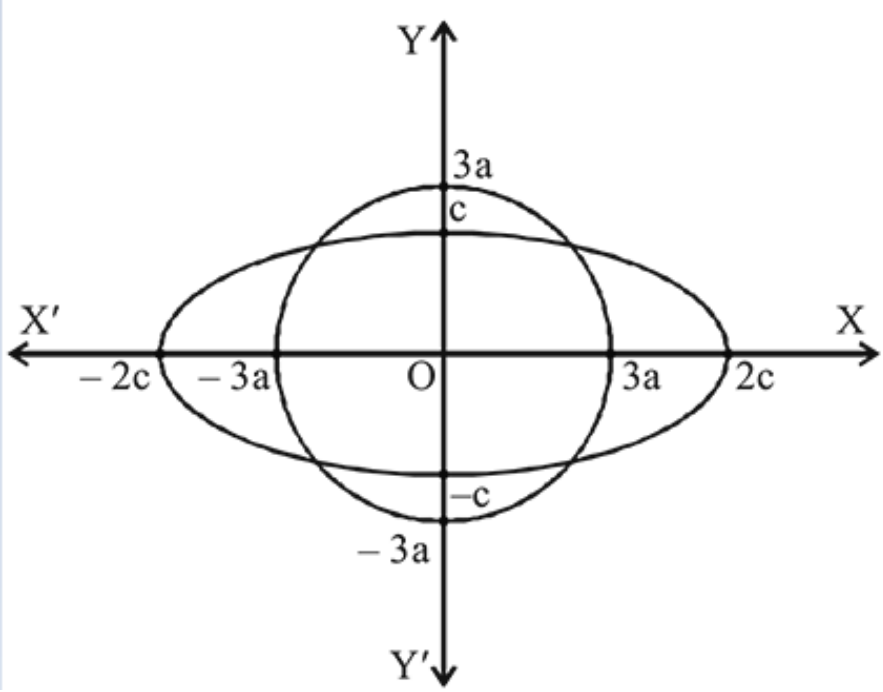
Also $b^2=a^2\left(1-e^2\right)=a^2-a^2e^2=a^2-25$

$\therefore$ **The equation (i) becomes** $\frac{(x-1)^2}{a^2}+\frac{(y-2)^2}{a^2-25}=1$

The point $(4,6)$ **lies on it, therefore** $\frac{(4-1)^2}{a^2}+\frac{(6-2)^2}{a^2-25}=1 \Rightarrow a^2=45$

$\therefore \mathrm{b}^2=45-25=20$

The required equation is $\frac{(x-1)^2}{45}+\frac{(y-2)^2}{20}=1$



$9ac-9a^2-2c^2>9ac-6c^2$

Again $3a<2c \Rightarrow 9ac<6c^2$

$\Rightarrow 9ac-6c^2<0$

From (i) and (ii), $9ac-9a^2-2c^2>0$

Q.4 Answer:

$\frac{1}{\sqrt{2}}$

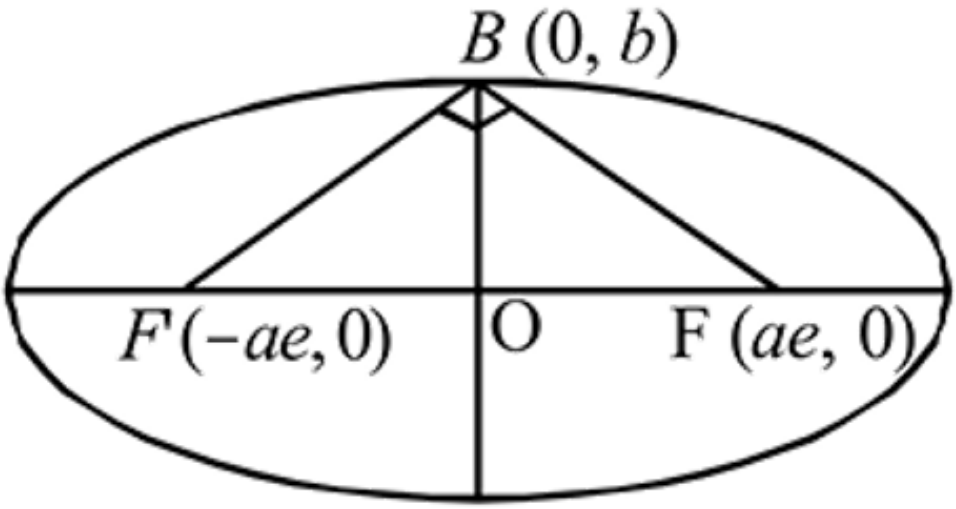
Solution:

$$(\mathrm{a}) \ \because \angle FBF'=90^\circ \Rightarrow FB^2+F'B^2=FF'^2$$

$$\therefore \left(\sqrt{a^2e^2+b^2}\right)^2+\left(\sqrt{a^2e^2+b^2}\right)^2=(2ae)^2$$

$$\Rightarrow 2\left(a^2e^2+b^2\right)=4a^2e^2 \Rightarrow e^2=\frac{b^2}{a^2} \cdots (\mathrm{i})$$

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Also, $e^2 = 1 - b^2/a^2 = 1 - e^2$ **(By using equation (i))**

$\Rightarrow 2e^2 = 1 \Rightarrow e = \frac{1}{\sqrt{2}}$

Q.5 Answer:

10

Solution:

The ellipse can be written as, $\frac{x^2}{25} + \frac{y^2}{16} = 1$ **Here** $a^2 = 25, b^2 = 16$**, but** $b^2 = a^2\left(1 - e^2\right) \Rightarrow$

$16/25$

$= 1 - e^2$

$\Rightarrow \quad e^2 = 1 - 16/25 = 9/25 \Rightarrow e = 3/5$

Foci of the ellipse are $(\pm ae, 0) = (\pm 3, 0)$**, i.e.,** F_1 **and** F_2

$\therefore$ **We have** $PF_1 + PF_2 = 2a = 10$ **for every point** P **on the ellipse.**

Q.6 Answer:

40

Solution:

Given eq ^{n} **of ellipse is** $\frac{x^2}{25} + \frac{y^2}{16} = 1 \Rightarrow a = 5$ **and** $b = 4$

Since $p(x, y)$ **moves along ellipse and** C **is the center**

$\max(\mathrm{CP}) = 5$ **and** $\min(\mathrm{CP}) = 4$

$\therefore 4\,\mathrm{Max}\{\mathrm{cp}\} + 5\,\min(\mathrm{cp}) = 4 \times 5 + 5 \times 4 = 40$

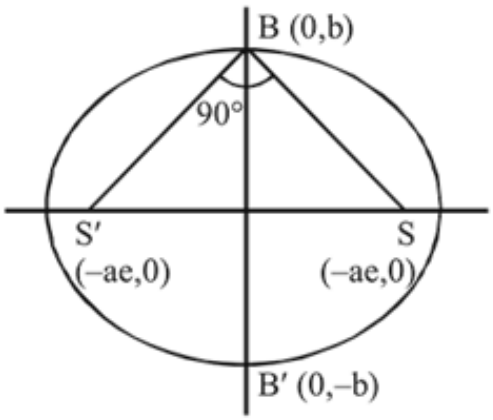
Q.7 Answer:

$\frac{1}{\sqrt{2}}$

Solution:

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From standard equation of ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, the co-ordinates of foci are



$\mathrm{S}(\mathrm{ae}, 0)$ **and** $\mathrm{S}'(-\mathrm{ae}, 0)$.

Co-ordinate of an extremity of the minor axis is $\mathrm{B}(0,\ \mathrm{b})$

Now slope of straight line $\mathrm{BS}=\frac{\mathrm{b}-0}{0-\mathrm{ae}}=\frac{\mathrm{b}}{-\mathrm{ae}}=\mathrm{m}_1$

and slope of straight line

$\mathrm{BS}'=\frac{\mathrm{b}-0}{0-(-\mathrm{ae})}=\frac{\mathrm{b}}{\mathrm{ae}}=\mathrm{m}_2$

$\because \mathrm{SB} \perp \mathrm{BS}'$, **so** $\mathrm{m}_1 \cdot \mathrm{m}_2=-1$ **or**

$\frac{\mathrm{b}}{-\mathrm{ae}} \times \frac{\mathrm{b}}{\mathrm{ae}}=-1$; **or** $\frac{\mathrm{b}^2}{\mathrm{a}^2}=\mathrm{e}^2$

But

$\mathrm{b}^2=\mathrm{a}^2\left(1-\mathrm{e}^2\right)$; **or** $\frac{\mathrm{b}^2}{\mathrm{a}^2}=1-\mathrm{e}^2$; **or** $\mathrm{e}^2=1-\mathrm{e}^2$

or $2\mathrm{e}^2=1$; **or** $\mathrm{e}=\frac{1}{\sqrt{2}}$.

Q.8 Answer:

an ellipse

Solution:

$\therefore (\alpha-5)^2+(\alpha)^2=(\alpha)^2$

$\Rightarrow \alpha^2+25-10\alpha=0 \Rightarrow (\alpha-5)^2=0 \Rightarrow \alpha=5$

Then, other root will always real.

Q.9 Answer:

$\frac{x^2}{k^2}+\frac{y^2}{k^2-h^2}=1$

Solution:

Let the equation of the ellipse be $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

Let e be the eccentricity of the ellipse. Since distance between foci $=2\ \mathrm{h} \therefore \quad 2\mathrm{ae}=2\ \mathrm{h} \Rightarrow \mathrm{ae}=\mathrm{h}$

Focal distance of one end of minor axis say $(0,\ \mathrm{b})$ **is** k

$\therefore \quad a+e(0)=k \Rightarrow a=k$

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From (1) and (2), $b^2 = a^2\left(1 - e^2\right) = k^2 - h^2$

$\therefore$ **The equation of the ellipse is**

$\frac{x^2}{k^2} + \frac{y^2}{k^2 - h^2} = 1$

Q.10 Answer:

$\frac{\sqrt{3}}{2}$

Solution:

Length of latus rectum of an ellipse is $\frac{2b^2}{a}$ **where** b **is semi minor axis and a is semi-major axis. As given,** $\frac{2b^2}{a} = b$

$\Rightarrow \quad 2\,\mathrm{b} = \mathrm{a} \Rightarrow \frac{\mathrm{b}}{\mathrm{a}} = \frac{1}{2}$

We know that eccentricity $\mathrm{e} = \sqrt{1 - \frac{\mathrm{b}^2}{\mathrm{a}^2}}$

$= \sqrt{1 - \frac{1}{4}} = \frac{\sqrt{3}}{2}$

Q.11 Answer:

6 units

Solution:

The sum of focal distances from a point of ellipse is 2 times the major axis.

For the given ellipse, length of semi major axis i.e. b is 3.

So required length $= 2 \times 3 = 6$

Option B is true

Q.12 Answer:

$\left(\frac{2\sqrt{91}}{7}, \frac{3\sqrt{105}}{14}\right)$

Solution:

Let the point is $(4\cos\theta, 3\sin\theta)$

According to question, $(4\cos)^2 + (3\sin\theta)^2 = \left(\frac{4+3}{2}\right)^2$

(1) $16 - 7\sin^2\theta = \frac{49}{4} \Rightarrow \sin^2\theta = \frac{15}{28}$

From

$\therefore \sin\theta = \pm\frac{1}{2}\sqrt{\frac{15}{7}} = \pm\frac{\sqrt{105}}{14}$

Similarly, $\cos\theta = \pm\frac{\sqrt{91}}{14}$

So the points are $\left(\frac{2\sqrt{91}}{7}, \frac{3\sqrt{105}}{14}\right)$;

$\left(-\frac{2\sqrt{91}}{7}, -\frac{3\sqrt{105}}{14}\right)$

Interchange θ **by** $\frac{\pi}{2} + \theta$ **and** $\frac{3\pi}{2} + \theta$.

Q.13 Answer:

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decreases with increase in λ

Solution:

Equation of the given conic is an equation of ellipse

$$\frac{x^2}{a^2+\lambda}+\frac{y^2}{b^2+\lambda}(x \geq 0)$$
$$\Rightarrow \quad A^2=a^2+\lambda \text{ and } B^2=b^2+\lambda$$
$$\text{Eccentricity, } e=\sqrt{1-\frac{B^2}{A^2}}=\sqrt{1-\frac{b^2+\lambda}{a^2+\lambda}}$$
$$=\sqrt{\frac{a^2+\lambda-b^2-\lambda}{a^2+\lambda}}$$
$$=\sqrt{\frac{a^2-b^2}{a^2+\lambda}}$$

λ is in the denominator so, when λ increases, the eccentricity decreases.

Q.14 Answer:

$3x^2+4y^2=300$

Solution:

Let the equation of the ellipse be $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. **If** S **and** S′ **be the foci, then** $\mathrm{SS}'=10$. **But** $\mathrm{SS}'=\mathrm{CS}+\mathrm{CS}'=2\mathrm{ae}$, C **being the centre**

$\therefore 2\mathrm{ae}=10$, **or** $\mathrm{ae}=5$

Also $2\frac{\mathrm{b}^2}{\mathrm{a}}=15$

Also $b^2=a^2\left(1-e^2\right)=a^2-a^2e^2=a^2-25$[**using** (1)]

$\mathrm{By}(2), 2\,\mathrm{b}^2=15\mathrm{a}$; **or** $2\left(\mathrm{a}^2-25\right)=15\mathrm{a}$

$\therefore \mathrm{a}=-\frac{5}{2}$ **or** $\mathrm{a}=10$

But a cannot be negative, $\therefore \mathrm{a}=10; \quad \therefore \mathrm{b}^2=\frac{15\times 10}{2}=75$

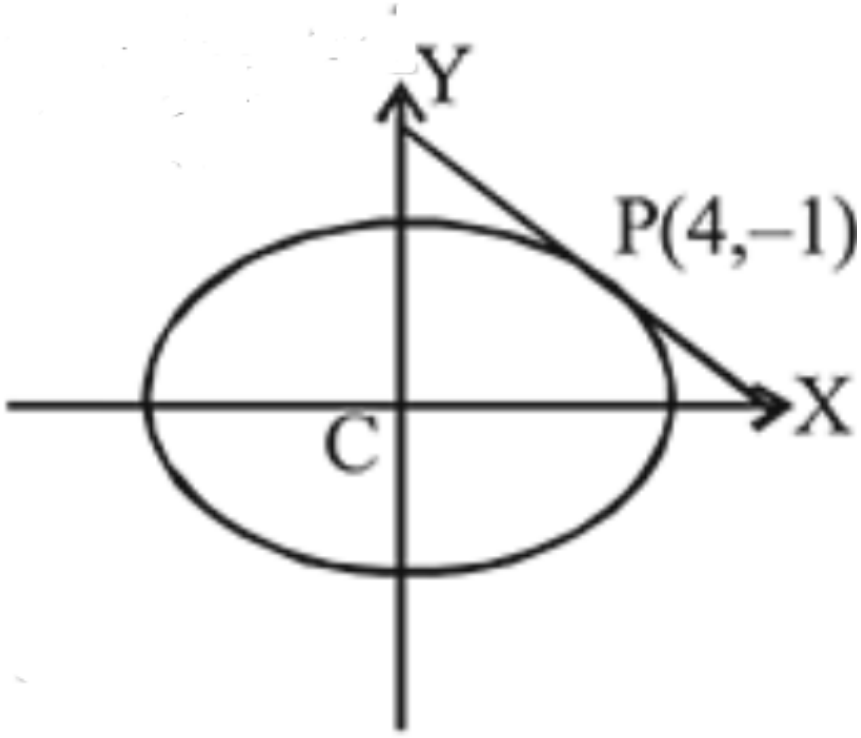
The equation to the ellipse is therefore $\frac{x^2}{100}+\frac{y^2}{75}=1$; **or** $3x^2+4y^2=300$

Q.15 Answer:

Both (a) and (b)

Solution:

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Let the equation of ellipse be $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ **The point** $\mathrm{P}(4,-1)$ **lies on it**

$\therefore \quad \frac{16}{a^2}+\frac{1}{b^2}=1 \quad \ldots$ **(1)**

Equation of a tangent to the ellipse is

$x+4y-10=0 \Rightarrow y=-\frac{1}{4}x+\frac{5}{2}$

$\therefore \left(\frac{5}{2}\right)^2=a^2\left(-\frac{1}{4}\right)^2+b^2$

[**using** $c^2=a^2m^2+b^2$]

$\Rightarrow \frac{25}{4}=\frac{a^2}{16}+\frac{a^2}{a^2-16}$[**from** (1)]

$\Rightarrow a^4-100a^2+1600=0 \Rightarrow a^2=80$ **or a** $^2=20$

$\Rightarrow a=4\sqrt{5}$ **or** $a=2\sqrt{5}$

From equation (1)

If $a^2=80, b^2=\frac{5}{4}$ **and if** $a^2=20, b^2=5$

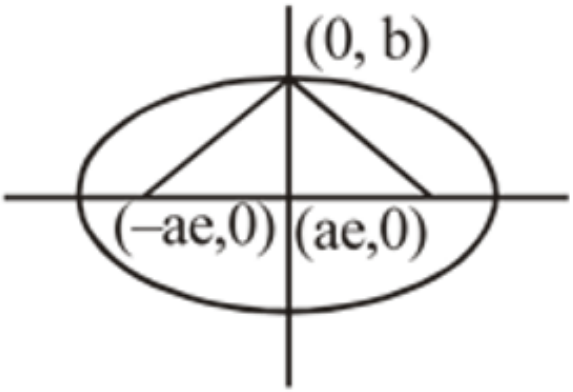
- Equation of the ellipse $\frac{x^2}{80}+\frac{y^2}{5/4}=1$ **or** $\frac{x^2}{20}+\frac{y^2}{5}=1$

Q.16 Answer:

$\frac{1}{\sqrt{2}}$

Solution:

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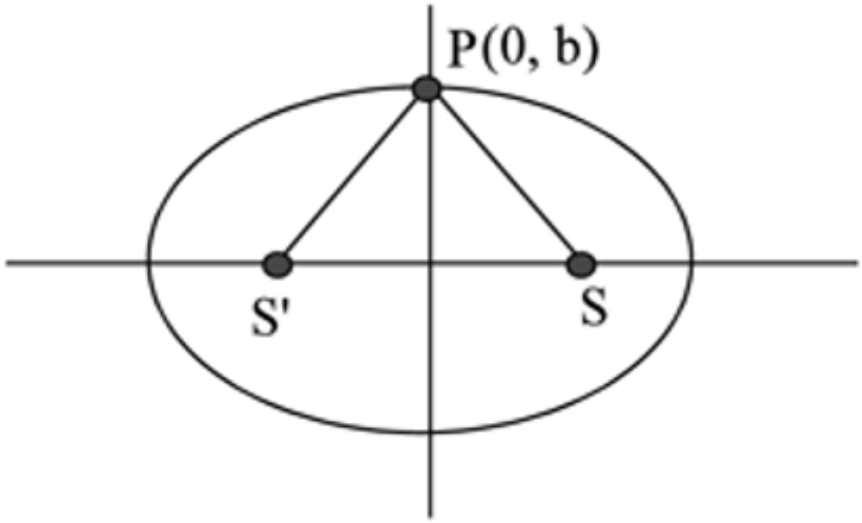


By Pythagoras theorem $2\left(b^2+a^2e^2\right)=4a^2e^2$
Hence $b^2/a^2=e^2$**. For an ellipse** b^2/a^2
$=1-e^2$ **which gives** $e=1/\sqrt{2}$

Q.17 Answer:

$\frac{1}{2}$

Solution:



$PS=SS^u\Rightarrow a=2ae\Rightarrow e=\frac{1}{2}$

Q.18 Answer:

15π

Solution:

It is given that the distance between two flags posts from him is always 10 meter hence its path is ellipse such that $2a=10\Rightarrow a=5$ **distance between two foci** $=2ae=8\Rightarrow e=\frac{4}{5}$
So $b^2=a^2\left(1-e^2\right)\Rightarrow b=3$
area $=\pi xb=15\pi$

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Q.19 Answer:

$2\sqrt{7}$

Solution:

Here $3x + 4y = 12\sqrt{2}$ is the tangent to ellipse.

$\Rightarrow y = -3x/4 + 3\sqrt{2} \ldots (1)$

We know, the equation of tangent to ellipse $x^2/a^2 + y^2/b^2 = 1$ is $y = mx + \sqrt{\left(a^2m^2 + b^2\right.}$

Here $b^2 = 9$

So, equation of tangent to given ellipse is $y = mx + \sqrt{(a^2m^2 + 9)}$

Now, using (1), we have

$m = -3/4$ and $\sqrt{(a^2m^2 + 9)} = 3\sqrt{2}$

$\Rightarrow \left(a^2m^2 + 9\right) = 18$

$\Rightarrow \left(a^2(-3/4)^2 + 9\right) = 18$

$\Rightarrow a^2 \times 9/16 = 9$

$\Rightarrow a = 4$

Now, $e^2 = 1 - b^2/a^2 = 1 - 9/16 = 7/16$

Or $e = \sqrt{7/4}$

The distance between foci is $2ae = 2 \times 4 \times \sqrt{7/4} = 2\sqrt{7}$

Q.20 Answer:

$5x^2 - 2xy + 5y^2 - 76x - 88y + 506 = 0$

Solution:

By definition, $SP = e$.

$PM = \sqrt{(x-6)^2 + (y-7)^2} = \frac{1}{\sqrt{3}} \cdot \frac{x+y+2}{\sqrt{1+1}}$

Squaring on both side,

$6\left(x^2 + y^2 - 12x - 14y + 85\right) = (x + y + z)^2$

or $5x^2 + 5y^2 - 76x - 88y - 2xy + 506 = 0$

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